

TEST 3

Course: Applied Statistics for Engineering and Scientists 1 **Course ID:** STAT451

Duration: 90 minutes

Materials: One A4 formula sheet allowed

Student name: **Student ID:** **No.:**

Note:

- All calculations must be rounded to **four** decimal places.
- Standard normal $\mathcal{N}(0, 1)$ and Student t distribution tables are provided.

Question 1. (2.0 *points*) In the production of a certain pharmaceutical tablet, the active ingredient content (in mg) is approximately normally distributed with mean $\mu = 250$ mg and standard deviation $\sigma = 6$ mg. Tablets are considered acceptable only if the active-ingredient content is between 240 mg and 260 mg.

- (a) Compute the probability that a randomly chosen tablet is *rejected* (i.e., outside specification).
- (b) A large batch of 200 tablets is randomly selected from production. Using the normal approximation to the binomial distribution, calculate the probability that the batch contains between 15 and 25 rejected tablets (inclusive).
- (c) A quality inspector randomly selects $n = 40$ tablets. Determine the probability that the sample mean active-ingredient content exceeds 252 mg.

Question 2. (2.0 *points*) The impact strength (in foot-pounds) of insulating materials used in printed-circuit boards is being monitored. Historical records indicate that impact strength is approximately normally distributed with population standard deviation $\sigma = 0.45$ foot-pounds. A random sample of $n = 36$ specimens produced a sample mean of $\bar{x} = 1.58$ foot-pounds.

- (a) Construct a 99% confidence interval for the true mean impact strength μ .
- (b) In a separate inspection, 42 out of 180 randomly selected circuit boards failed a thermal-shock test. Construct a 90% confidence interval for the true proportion p of boards that fail the test.
- (c) The quality engineer desires a 95% confidence interval for μ with a maximum error of 0.10 foot-pounds. What minimum sample size is required?

Question 3. (2.0 *points*) A brochure claims that the mean tensile strength of a certain synthetic fiber is 84 cN/tex. To test this, a quality control laboratory measures a random sample of $n = 12$ fibers. Assuming the tensile strength is normally distributed, the sample yielded $\bar{x} = 82.1$ cN/tex and $s = 3.4$ cN/tex.

- (a) At the significance level $\alpha = 0.05$, test $H_0 : \mu = 84$ against $H_1 : \mu < 84$. Clearly state the rejection region, the value of the test statistic, and the conclusion in plain English.

- (b) A field study involving 500 fibers found that 85 of them broke under the specified working load. The manufacturer claims that the true breakage rate is less than 20%. Test this claim at $\alpha = 0.05$.
- (c) Compute the p -value for part (b) and interpret its meaning in context.

Question 4. (1.5 points) A dietician wishes to evaluate a new weight-loss supplement. Ten randomly selected volunteers were weighed (in kilograms) before starting the program and again after 8 weeks.

Subject	1	2	3	4	5	6	7	8	9	10
Before	86.4	79.2	92.1	74.8	88.5	81.0	95.3	77.6	83.9	90.7
After	83.1	77.8	88.4	73.5	85.2	79.6	91.7	76.9	81.0	86.8

Assume the differences are approximately normal.

- (a) Test at $\alpha = 0.05$ whether the supplement produces a mean weight loss of more than 2 kg. Use the paired t -test and state all steps.
- (b) Construct a 95% confidence interval for the mean weight loss μ_D .

Question 5. (2.5 points) The final exam score (y) and the number of hours spent studying per week (x) are recorded for $n = 9$ randomly chosen students in an introductory statistics course:

x	2	3	4	6	7	8	10	12	14
y	55	58	63	70	72	78	83	88	92

- (a) Determine the least-squares regression line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. Interpret the estimated slope and the intercept in context.
- (b) Compute the sample correlation coefficient r between study hours and exam score, and interpret its value.
- (c) Compute the coefficient of determination R^2 and interpret it.
- (d) Using the regression line, predict the exam score for a student who studies 9 hours per week and for a student who studies 25 hours per week. Comment on the reliability of each prediction.

— END OF TEST 3 —